

Role of Quantitative EEG and EEG Reactivity in Traumatic Brain Injury

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Abstract

Objective. This study aimed to explore the effectiveness of quantitative electroencephalogram (EEG) and EEG reactivity (EEG-R) to predict the prognosis of patients with severe traumatic brain injury. **Methods.** This was a prospective observational study on severe traumatic brain injury. Quantitative EEG monitoring was performed for 8 to 12 hours within 14 days of onset. The EEG-R was tested during the monitoring period. We then followed patients for 3 months to determine their level of consciousness. The Glasgow Outcome Scale (GOS) score was used. The score 3, 4, 5 of GOS were defined good prognosis, and score 1 and 2 as poor prognosis. Univariate and multivariate analyses were employed to assess the association of predictors with poor prognosis. **Results.** A total of 56 patients were included in the study. Thirty-two patients (57.1%) awoke (good prognosis) in 3 months after the onset. Twenty-four patients (42.9%) did not awake (poor prognosis), including 11 cases deaths. Univariate analysis showed that Glasgow coma scale (GCS) score, the amplitude-integrated EEG (aEEG), the relative band power (RBP), the relative alpha variability (RAV), the spectral entropy (SE), and EEG-R reached significant difference between the poor-prognosis and good-prognosis groups. However, age, gender, and pupillary light reflex did not correlate significantly with poor prognosis. Furthermore, multivariate logistic regression analysis showed that only RAV and EEG-R were significant independent predictors of poor prognosis, and the prognostic model containing these 2 variables yielded a predictive performance with an area under the curve of 0.882. **Conclusions.** Quantitative EEG and EEG-R may be used to assess the prognosis of patients with severe traumatic brain injury early. RAV and EEG-R were the good predictive indicators of poor prognosis.

Keywords

quantitative EEG, EEG reactivity, prognosis, traumatic brain injury

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Introduction

Traumatic brain injury (TBI) is the major cause of disability and mortality worldwide. The management of severe traumatic brain injury (sTBI) is full of challenges, and the assessment of patients is particularly important early. Electroencephalogram (EEG) reflects the brain function directly and sensitively. Thus, it was recommended for the use in comatose patients.¹ But the raw EEG is complicated and difficult to explain for nonneurologists. Quantitative EEG (qEEG) is a technique for the quantification of the raw EEG by means of mathematical and computerized methods and the quantized EEG data are obtained and presented in the form of a trend chart. These data are easier to understand compared with the raw EEG. Studies have shown that qEEG may be used as an effective evaluation method for patients with acute stroke, subarachnoid hemorrhage, hypoxic ischemic encephalopathy, and neonatal patients intensive care unit.^{2–7} However, studies on sTBI have less been reported. EEG reactivity (EEG-R) is defined as a change in either the background frequency or amplitude following stimulations.⁸ EEG-R

has a certain predictive value for comatose patients, and experts recommended EEG-R as an indicator to predict the prognosis of comatose patients.^{9–11} However, studies on the predictive performance of EEG-R in comatose patients were not conclusive. In the following study, we assessed the patients of sTBI by combining qEEG with EEG-R, aiming to find an effective method for the prognosis of patients early.

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Material and Methods

Patients

A prospective study was conducted on consecutive patients with sTBI admitted from February 2017 to June 2018 to neurosurgery intensive care unit (N-ICU) and general ICU at Xiangya Hospital of Central South University. The main treatment principles referred to the *Guidelines for the Management of Severe Traumatic Brain Injury* 4th edition by American Brain Trauma Foundation in 2016. All subjects signed a written informed consent form, and the study did not involve changes of the routine medical practices. The inclusion criteria were as follows: All patients older than 18 years having a Glasgow Coma Scale (GCS) score of less than or equal to 8, and the onset of brain injury within 14 days. Patients with mild hypothermia therapy required to rewarm to above 35 °C.^{12,13} The exclusion criteria were as follows: (1) a known history of neurological disorders; (2) EEG recorded during sedation maintained by greater than 2 mg/h of midazolam or the infusion dose of propofol higher than 1.2 mg/kg/h¹⁴; (3) burst-suppression; and (4) patients lost to follow-up.

qEEG Recording

Bedside qEEG (Nicolet Monitor, Natus Inc) was performed within 14 days of onset. EEG monitoring was mostly performed at night because of fewer operations. The filter channel was set at 1 to 35 Hz, the sample frequency at 125 Hz, and time constant at 0.3 seconds. The electrodes were placed according to the international 10-20 system, using 8 channels (Fp1, Fp2, C3, C4, T3, T4, O1, and O2). Cz and Fpz were considered as reference and ground, respectively. Bipolar longitudinal leads were used in qEEG recording (Fp1-T3, T3-O1, Fp1-C3, C3-O1, Fp2-T4, T4-O2, Fp2-C4, and C4-O2). All patients were monitored for 8 to 12 hours, and tube bed nurse helped supervise EEG.

qEEG Parameters

Amplitude-integrated electroencephalogram (aEEG), relative band power (RBP), relative alpha variability (RAV), and spectral entropy (SE) were acquired through raw EEG (Figure 1). aEEG was a qEEG parameter based on raw EEG amplitude. The maximum amplitude of EEG within 15 seconds constituted the upper bound of aEEG, and the minimum amplitude constituted the lower bound of aEEG. aEEG was divided into 3 grades: grade I, normal amplitude (aEEG upper edge >10 μ V and aEEG lower edge >5 μ V); grade II, mild abnormal amplitude (upper edge >10 μ V and lower edge $\leq$ 5 μ V, or upper edge $\leq$ 10 μ V and lower edge >5 μ V); and grade III, suppressed amplitude (upper edge <10 μ V and lower edge <5 μ V).¹⁵ the lower the aEEG value, the higher the grade. RBP was the percentage of each frequency band power of EEG in the total band power. RBP data were calculated every 10 seconds, and different colors represented different band powers: red, delta frequency (1-4 Hz); yellow, theta frequency (4-8 Hz); green, alpha

frequency (8-14 Hz); and blue, beta frequency (14-30 Hz). SE was the EEG power spectrum obtained by Fourier transformation based on frequency domain analysis. SE values were calculated every 10 seconds and presented in the form of a line graph. The relative alpha (RA) data were calculated every 2 minutes automatically. The frequency band for RA was the proportion of 6 to 14 Hz frequency power in 1 to 20 Hz total frequency power activity.³ These RA trends were collected and then scored for their variability (RAV) by a physician blinded to the clinical condition and patient outcomes. The RAV is calculated using the formula: (peak RA – trough RA)/(peak RA + trough RA).³ An average baseline of the RA trend was set at a median value of RA during 8 to 12 hours. Excursions once per hour greater than 15% of the median values were recorded as excellent (grade 4); excursions of 10% at least every 4 hours were recorded as good (grade 3); small excursions during the whole monitoring process recorded as fair (grade 2); and excursions less than 2% of the median values were recorded as poor (grade 1).^{3,16} Ultimately, the RAV was scored as excellent, good, fair, or poor (Figure 2).

EEG-R test

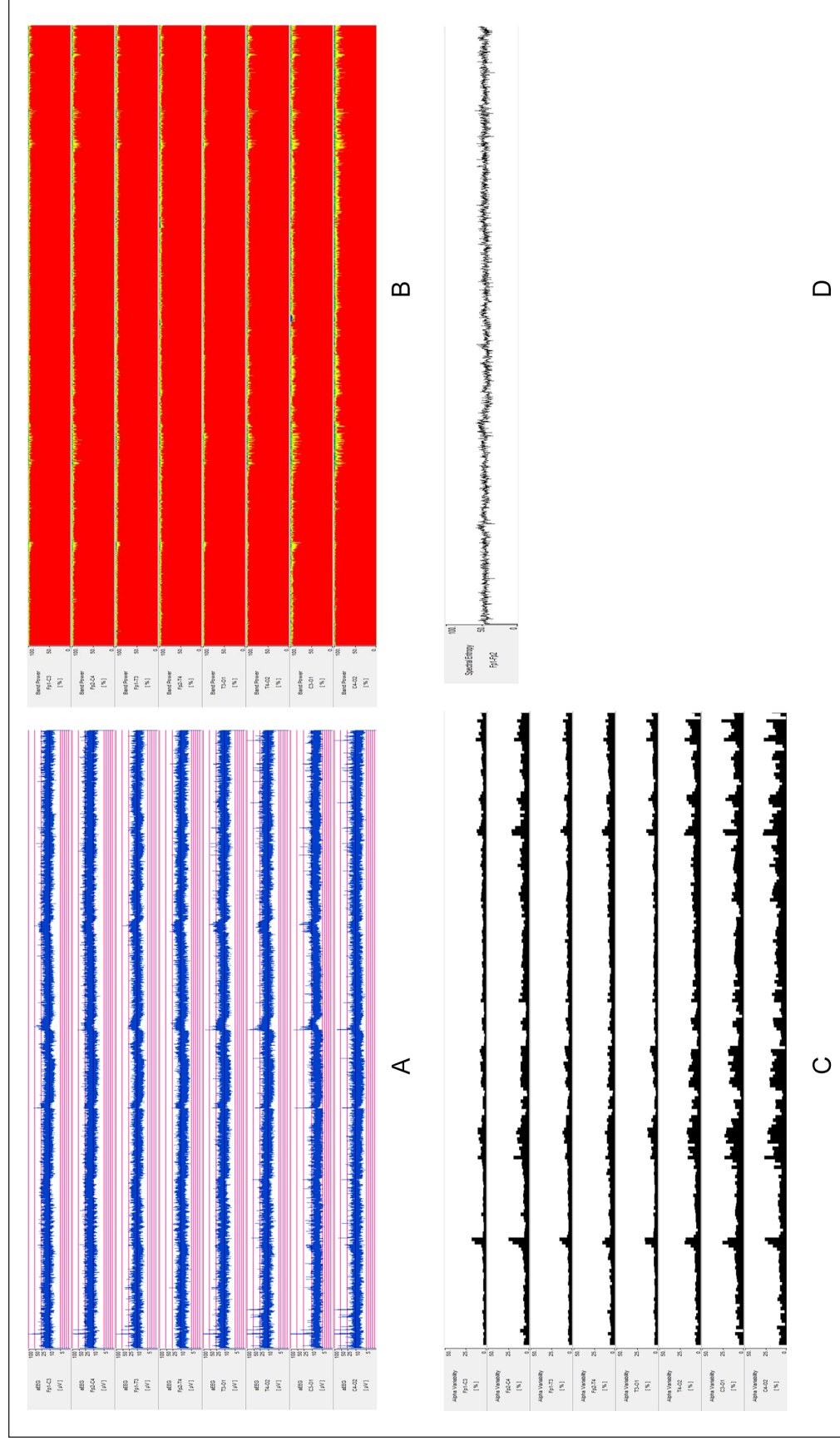
EEG reactivity testing was executed through nail bed pressure and clapping, calling out the patient's name. Each stimulus repeated at least 3 times with duration for longer than 5 seconds.¹¹ EEG-R was assessed by 2 certified neurologists (certified by Brain Injury Evaluation Quality Control Centre of China) separately. The definition of EEG-R is "A reproducible change (at least twice) in the EEG in response to stimulation." EEG-R was present if any change in the EEG, visible with the naked eye. The EEG-R was classified as reactive (present) or nonreactive (absent), as shown in Figure 3.

Predictor Variables and Outcome Definitions

We based our clinical variables on available observation in all selected patients. As a result, the following variables were assessed for each patient: age, gender, GCS score, pupillary light reflex on admission to the ICU, aEEG, RBP, RAV, SE, and EEG-R. The upper and lower bounds of aEEG, RBP, and RAV values, which were the average values of the bipolar longitudinal monitoring channels, were obtained. SE data were the frontal pole monitoring channel values. The outcomes of all patients were assessed for 3 months after entry into the study. The Glasgow Outcome Scale (GOS) score (5, good cerebral performance; 4, moderate cerebral disability; 3, severe cerebral disability; 2, vegetative state (VS); and 1, death) was used.¹⁷ The scores 3, 4, and 5 of GOS were defined as awakened and good prognosis, and score 1 and 2 as unawakened and poor prognosis.

Statistical Analyses

SPSS software, version 22 (IBM Corp) was used for all statistical analyses. We performed chi-squared tests for count variables,



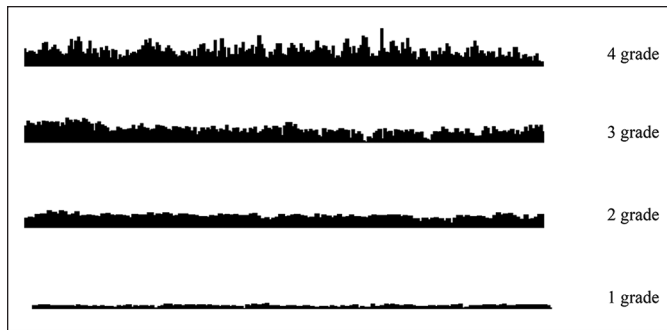


Figure 2. Relative alpha variability (RAV) grades (grade 1, poor; grade 2, fair; grade 3, good; grade 4, excellent).

and 2-tailed *t* tests for normally distributed continuous variables were used. continuous normally distributed variables were expressed as mean $\pm$ standard deviation (SD) and categorical variables were expressed as counts and percentages. If measurement data were not normally distributed, Mann-Whitney *U* test was used, the data were described by the median (interquartile range). The predictors of poor prognosis were assessed with the area under the receiving operator characteristic (ROC) curve, and they were assessed for sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and unweighted accuracy. Furthermore, logistic regression analysis was used to analyze these parameters. A *P* value $< .05$ was considered as statistically significant.

Results

A total of 67 patients with sTBI were enrolled in this study. Eleven patients had to be excluded: 2 patients died of severe infection, 3 patients were lost to follow-up, 2 patients appeared burst suppression, and 4 patients presented epileptic status. Therefore, 56 patients were included in this study, of which 36 (64.3%) were males. The mean (SD) age was 54.8 (11.9) years. The median time of EEG monitoring was the fourth (interquartile range, IQR 3-10) day after onset. After 3 months, 32 patients (57.1%) had a good prognosis: 5 good recovery, 10 moderate disabilities, 17 severe disabilities. 24 patients (42.9%) had a poor outcome: 13 patients vegetative states and 11 patients deaths (Table 1). The activity of delta frequency was dominant in patients with sTBI, while the activity of alpha frequency decreased significantly. According to the Synek classification standard,¹⁸ The EEG of most patients was at grade III and 5 patients were at grade II. Four patients were at grade IV, of whom 3 patients were low voltage and 1 patient was alpha coma. The clinical and qEEG parameters collected at the start of the study in awakened and nonawakened patients were summarized in Table 2. The values of aEEG upper and lower bounds in the poor-prognosis group were lower compared with that in the good-prognosis group ($P = .01$). aEEG was divided into 3 grades based on the values of upper and lower bounds. Thirty-two patients with aEEG grade I, 16 patients with aEEG grade II, and 8 patients with aEEG grade III (Table 1). When

aEEG was at grade I, 23 patients awoke, while only 1 out of 8 patients awoke when aEEG was at grade III. The delta relative band power in the poor-prognosis group was significantly higher than that in the good-prognosis group ($P = .001$), while the alpha relative band power, the beta relative band power and the theta relative band power in the good-prognosis group were higher than that in the poor-prognosis group, and then the differences were all significant statistically. The mean spectral entropy of the poor-prognosis group was lower than that of the good-prognosis group, and the significant difference between the 2 groups was reached ($P = .008$). When EEG-R was present, 23 out of 29 patients awoke, while only 9 out of 27 patients awoke when it was absent. When RAV was at grade 3 to 4, 27 out of 31 patients awoke, while only 5 of 25 patients awoke when RAV was at grade 1 to 2. Univariate analysis showed that patients were more likely to awaken if they presented with EEG-R, RAV grade 3 to 4 and higher admission GCS scores. In the contrast, age ($P = .358$), gender ($P = .809$), pupillary light reflex ($P = .053$) did not correlate significantly with 3 months awakening.

The receiver operating characteristic (ROC) curves were based on the sensitivity and specificity of significant variables in the prediction of poor prognosis and are shown in Figure 4. The sensitivity of RAV for poor prognosis was 83.3%, the specificity was 84.4%, and the area under the ROC curves (AUC) was 0.839; Similarly, the sensitivity of EEG-R was 75%, the specificity was 71.9%, and the ROC-AUC was 0.734, which were superior to GCS score, as shown in Table 3.

Furthermore, variables that were significantly different between the 2 groups in univariate analysis (GCS, aEEG, RBP, RAV, SE, and EEG-R) were entered into multivariate regression analysis. The results demonstrated that only RAV and EEG-R were significantly different with the prediction of poor prognosis. The regression model showed a good predictive performance with an AUC of 0.882 (Table 4).

Discussion

The aim of the study was to establish relationships between different clinical parameters and quantitative EEG with clinical outcomes. Quantitative EEG was a useful tool in traumatic brain injury.¹⁹⁻²¹ In the study, traffic accidents and fall injuries were the causes of TBI, most of which were closed blunt brain injuries. Notably, TBI caused by gunshots are very rare in China because China has the strictest gun control laws worldwide.²²

In the study, we excluded patients with deep sedation, and the study did not affect the treatment. Unless the patient had a seizure during the monitoring process, we would intervene with large doses of medications and the patient withdrew from the study. Among the clinical parameters collected, we found that the combination of qEEG and EEG-R presented good predictive performance for poor prognosis. Furthermore, we developed a predictive model containing RAV and EEG-R that showed good discriminative power with an AUC of 0.882 for



Figure 3. EEG reactivity (EEG-R). (A) Frequency and amplitude changes after stimulation, indicating the presence of EEG-R. (B) No change of EEG after stimulation, indicating the absence of EEG-R.

Table 1. The Characteristics of the 56 Patients.

Characteristic	Value
Age, years, mean $\pm$ SD	54.8 $\pm$ 11.9
Gender, male/female, n	36/20
Time from onset to EEG recording, days, median (IQR)	4 (3-10)
aEEG grade, n (%)	
Grade I	32 (57.1)
Grade II	16 (28.6)
Grade III	8 (14.3)
Outcome at 3 months, n (%)	
Good prognosis	32 (57.1)
Poor prognosis	24 (42.9)
GOS score at 3 month, n (%)	
Good recovery	5 (8.9)
Moderate disability	10 (17.9)
Severe disability	17 (30.4)
Vegetative state	13 (23.2)
Death	11 (19.6)

Abbreviations: IQR, interquartile range; aEEG, amplitude-integrated electroencephalography; GOS, Glasgow Outcome Scale.

poor prognosis. Thus, the study provided a new method for clinicians to predict the prognosis of patients with sTBI.

GCS score has been widely used in clinic for assessing brain injury because of its easy operation. However, the application of GCS score in some items was not specific enough, which caused differences in the evaluation, and it should not be considered accurate.²³ The findings in this study were weakly correlative between GCS score and outcome. The positive predictive value of predicting the poor prognosis of patients with sTBI was poor.

The previous studies demonstrated that EEG was useful in predicting the prognosis of patients with brain injury.^{24,25} However, the standard EEG was complicated, and it was

difficult to interpret accurately. qEEG parameters containing aEEG, RBP, RAV, SE, were obtained by calculating and analyzing the frequency and amplitude of the raw EEG, which was easier to explain to nonneurologists. AEEG contains the amplitude information of EEG and reflects the severity of brain injury directly. AEEG value decreases obviously under conditions of brain moderate hypoxia.²⁶ RBP contains the frequency information and reflects cerebral ischemia in the early stage. Continuous delta wave slowing or decreased alpha wave indicates poor outcome.^{27,28} RAV detects delayed cerebral ischemia development in patients with subarachnoid hemorrhage early.²⁹ The decline in RAV indicates a reduction in the focal cerebral blood flow, and detecting vascular spasm caused by ischemia 7 hours earlier than the appearance of clinical symptoms, and 44 hours earlier than diagnosed using computed tomography.^{3,29} SE is considered to quantify the complexity of EEG signals. When patients are conscious, their brain activity is irregular and complex with high SE. When cerebral cortex is inhibited, brain activity becomes orderly with low SE. Thus, it can be seen that qEEG parameters were a useful monitoring method in comatose patients, and its abnormal changes were earlier than clinical symptoms, and even superior to neuroimaging. This study aimed to explore the application value of qEEG parameters in patients with sTBI because of less research in this field.

In the study, the trend chart of aEEG was divided into 3 grades according to the average value of aEEG upper and lower bounds. The higher the grade, the more severe the damage, and the less likely the patient awoke. EEG slow waves dominated in comatose patients, SE of patients in coma decreased obviously.³⁰ The study showed that the slow waves dominated and SE decreased in patients with sTBI. The findings were insufficient on the prognosis of patients with sTBI by RBP or SE, and subsequent investigations with a large sampling size are needed.

EEG changes are closely tied to cerebral blood flow.³¹ The fast wave gradually decreased with the decrease in cerebral

Table 2. Univariate Analysis.

Parameters	Good prognosis	Poor prognosis	P
Age, years, mean $\pm$ SD	56.03 $\pm$ 10.56	53.04 $\pm$ 13.58	.358
Male/female	21/11	15/9	.809
Time from injury onset to EEG record, days, median (IQR)	4 (3-7.75)	5 (2.25-10)	.653
Pupillary light reflex			.053
Present ^a	24	12	
Absent	8	12	
aEEG (μ V)			
Upper bound, mean $\pm$ SD)	13.37 $\pm$ 5.05	10.09 $\pm$ 3.80	.010
Lower bound, mean $\pm$ SD)	8.32 $\pm$ 2.98	6.36 $\pm$ 2.31	.010
RBP (%)			
Delta, median (IQR)	84.02 (77.74-89.28)	89.95 (86.46-93.72)	.001
Theta, median (IQR)	11.15 (5.72-14.38)	7.64 (3.83-9.74)	.003
Alpha, median (IQR)	3.82 (2.19-5.12)	1.78 (1.36-3.03)	<.001
Beta, median (IQR)	1.80 (1.02-2.56)	0.95 (0.72-1.41)	.001
SE, mean $\pm$ SD	45.06 $\pm$ 4.40	41.08 $\pm$ 6.35	.008
GCS score on admission			.020
3-5	10	15	
6-8	22	9	
RAV			<.001
Grade 1-2	5	20	
Grade 3-4	27	4	
EEG-R			.001
Present	23	6	
Absent	9	18	

Abbreviations: IQR, interquartile range; aEEG, amplitude-integrated electroencephalogram; EEG-R, EEG reactivity; GCS, Glasgow Coma Scale; RAV, relative alpha variability; RBP, relative band power; SE, spectral entropy.

^aAt least one side presence.

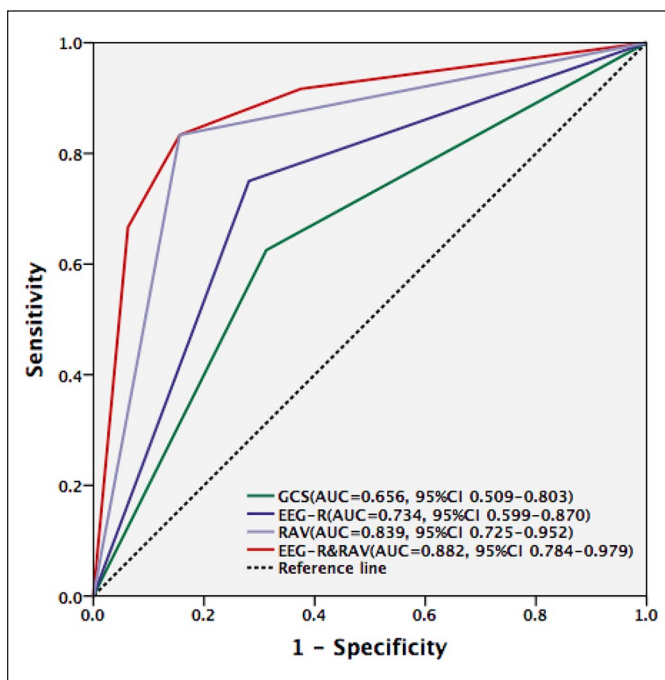


Figure 4. Receiver operating characteristic (ROC) curves showing the performance of prediction of poor prognosis.

blood flow, while slow wave increased gradually; RAV was gradually impaired and decreased in this process.^{29,31} A RAV value less than or equal to 0.1 (grades 1-2) highly predicted the poor prognosis of patient in TBI, with the sensitivity and specificity of more than 75%.¹⁴ RAV was divided into 4 grades in the study. The RAV value was more than 0.1 for grades 3 to 4, and it was less than or equal to 0.1 for grades 1 to 2.³ The study showed that the RAV value of patient with poor prognosis was more likely less than 0.1. The sensitivity of RAV in predicting poor prognosis of patient was 83.3%, the specificity was 84.4%, and the area under the curve of poor prognosis was 0.839. The results proved that RAV predicted the prognosis of patients with sTBI early.

EEG-R had a certain predictive value for the prognosis of comatose patients. It depends on the integrity of reticular structure and thalamocortical system, and represents the neural activity along the afferent somatosensory pathways from the ascending reticular activating system to the cortex after external stimulation.³² The presence of EEG-R in comatose patients indicates the recovery of consciousness, which suggests that the brain is responsive to the outside world.⁸ In the study, the sensitivity of EEG-R to the poor prognosis of patients with sTBI was 75%, which was superior to GCS score. However, the interpretation of EEG-R was still subjective.³³ Therefore,

Table 3. Performance of the Variables for Predicting Poor Prognosis.

	Se (%)	Sp (%)	PPV (%)	NPV (%)	Predictive accuracy (%)	AUC (95% CI)
GCS	62.5	68.8	60.0	71.0	66.1	0.656 (0.509-0.803)
RAV	83.3	84.4	80.0	87.1	83.9	0.839 (0.725-0.952)
EEG-R	75.0	71.9	66.7	79.3	73.2	0.734 (0.599-0.870)

Abbreviations: Se, sensitivity; Sp, specificity; PPV, positive predictive value; NPV, negative predictive value; AUC, area under the curve; GCS, Glasgow Coma Scale; RAV, relative alpha variability; EEG-R, EEG reactivity.

Table 4. Multivariable Logistic Regression Model for Predicting Poor Prognosis.

Variable	OR (95% CI)	P	AUC (95% CI)
RAV	19.477 (4.390-86.420)	<.001	0.882 (0.784-0.979)
EEG-R	4.197 (0.944-18.657)	.060	

Abbreviations: RAV, relative alpha variability; EEG-R, EEG reactivity; OR, odds ratio; AUC, area under the curve.

the evaluation of EEG-R was completed by 2 neurologists, and the inconsistent results would be settled through negotiation, aiming to avoid making mistakes.

The study envisaged combining qEEG with EEG-R to improve the accuracy of prediction. Using multivariate regression analysis, we established a prognostic model incorporating RAV with EEG-R and found that this model had outstanding performance in predicting poor prognosis from coma. The ROC-AUC of the model (0.882; 0.784-0.979) was superior to RAV and EEG-R. This suggested that a multimodal prognostication approach was best when determining coma patient outcomes.

The study had some limitations. First, the results obtained from the limited patients, and we did not provide data on 1-year outcome for the study. Some comatose patients may awake in one year. Second, the region of damage in patients was different, and only 8 channels were used for monitoring, which may not fully reflect the extent of brain injury. Third, sedation was potential confounder in the patients with sTBI, although the higher doses of continuously administrated sedatives were excluded. Fourth, qEEG parameters originated from the raw EEG, and the veracity of qEEG data was bound to decline when various artifacts appear in the raw EEG. We excluded interference values as much as possible when we analyzed and processed the data. However, there may still be some interference that was not identified. Furthermore, there may be errors on quantitative EEG itself.³⁴ Finally, the mechanism of brain injury in patients were not exactly the same. Given the relatively small sample size of this study, we did not further conduct the analysis. Future large sample size studies are needed.

In conclusion, Quantitative EEG and EEG-R may assess the prognosis of patients with sTBI early. Furthermore, RAV and EEG-R were the good predictive indicators of poor prognosis. Multicenter studies are still needed to confirm the conclusion.

Authors' Note

The ethics committee of Xiangya Hospital of Central South University approved the study (ethical approval number 201710998).

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Author Contributions

Jian Wang, Jinfang Liu, and Daomiao Xu contributed to the conception and design. Jian Wang and Li Huang performed the analysis and interpretation of data, and drafted manuscript together. All authors contributed to interpretation, revised manuscript critically and gave final approval. The authors agree to be accountable for all aspects of the work ensuring integrity and accuracy.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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